

Build & Flash Guide

This guide covers downloading the Flight Radar code and uploading it to your Waveshare ESP32-S3-Touch-LCD-1.46C board.

Quick Start (For the Impatient)

1. **Download:** <https://github.com/computethis/TheMakersSpace-Flight-Radar> → Code → Download ZIP
2. **Extract** to Desktop → Rename folder to **FlightRadar**
3. **Open:** **FlightRadar/src/FlightRadar/FlightRadar.ino** in Arduino IDE
4. **Install Libraries:** Arduino_GFX_Library (1.5.7+), ArduinoJson (7.x), WiFiManager (2.0.x), SensorLib
5. **Board Settings:** Select "Waveshare ESP32-S3-Touch-LCD-1.46", PSRAM: **Enabled**
6. **Upload** → Verify display shows "SETUP"

Stuck? See the detailed steps below.

Part 1: Download the Code

Option A: Download ZIP (Recommended)

1. Go to <https://github.com/computethis/TheMakersSpace-Flight-Radar>
2. Click the green "<> Code" button
3. Select "**Download ZIP**"
4. Extract the ZIP file to your Desktop
5. **Rename** the extracted folder to **FlightRadar** (removes the **-main** suffix)

You should now have:

```
Desktop/
├── FlightRadar/
│   ├── src/
│   │   └── FlightRadar/
│   │       ├── FlightRadar.ino    ← Open this
│   │       ├── board_config.h
│   │       ├── aircraft_types.h
│   │       ├── themes.h
│   │       ├── ui_layout.h
│   │       └── font_config.h
│   ├── docs/
│   └── README.md
```

Option B: Use Git (If Installed)

```
cd Desktop # or wherever you want to store the git repository
git clone https://github.com/computethis/TheMakersSpace-Flight-Radar.git
```

Part 2: Install Arduino IDE & ESP32 Support

First-Time Setup

If you already have Arduino IDE with ESP32 support, skip to Part 3.

1. Download Arduino IDE from <https://www.arduino.cc/en/software>
 - Version 1.8.x or 2.x both work
2. Install and launch Arduino IDE
3. Open **File** → **Preferences** (Arduino → Preferences on Mac)
4. Find "Additional Board Manager URLs" and add:
 1. https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
 - If there's already a URL there, click the icon and add this on a new line
5. Click **OK**
6. Go to **Tools** → **Board** → **Board Manager**
7. Search for "**ESP32**"
8. Install "**ESP32 by Espressif Systems**" (latest version)
9. Wait for installation to complete (may take a few minutes)

Part 3: Install Required Libraries

Go to **Sketch** → **Include Library** → **Manage Libraries...**

In the Library Manager search box, install each of these:

Library	Version	Author
GFX Library for Arduino	1.5.7 or higher	moon on our nation
ArduinoJson	7.x	Benoit Blanchon
WiFiManager	2.0.x	tzapu
SensorLib	Latest	Lewis He

Installation tips:

- Search for the exact name shown above
- Click **Install** for each library
- If you see multiple results, pick the one with the matching author

- Some libraries may already be installed— that's fine!

Shouldn't be needed, but some resources online recommend **Closing and Reopening Arduino IDE** after installing libraries.

Part 4: Open the Project

1. In Arduino IDE: **File** → **Open**
2. Navigate to `Desktop/FlightRadar/src/FlightRadar/`
3. Select `FlightRadar.ino`
4. Click **Open**

You should see:

- Main code in the editor
 - Tabs at the top: FlightRadar.ino, board_config.h, aircraft_types.h, themes.h, font_config.h
-

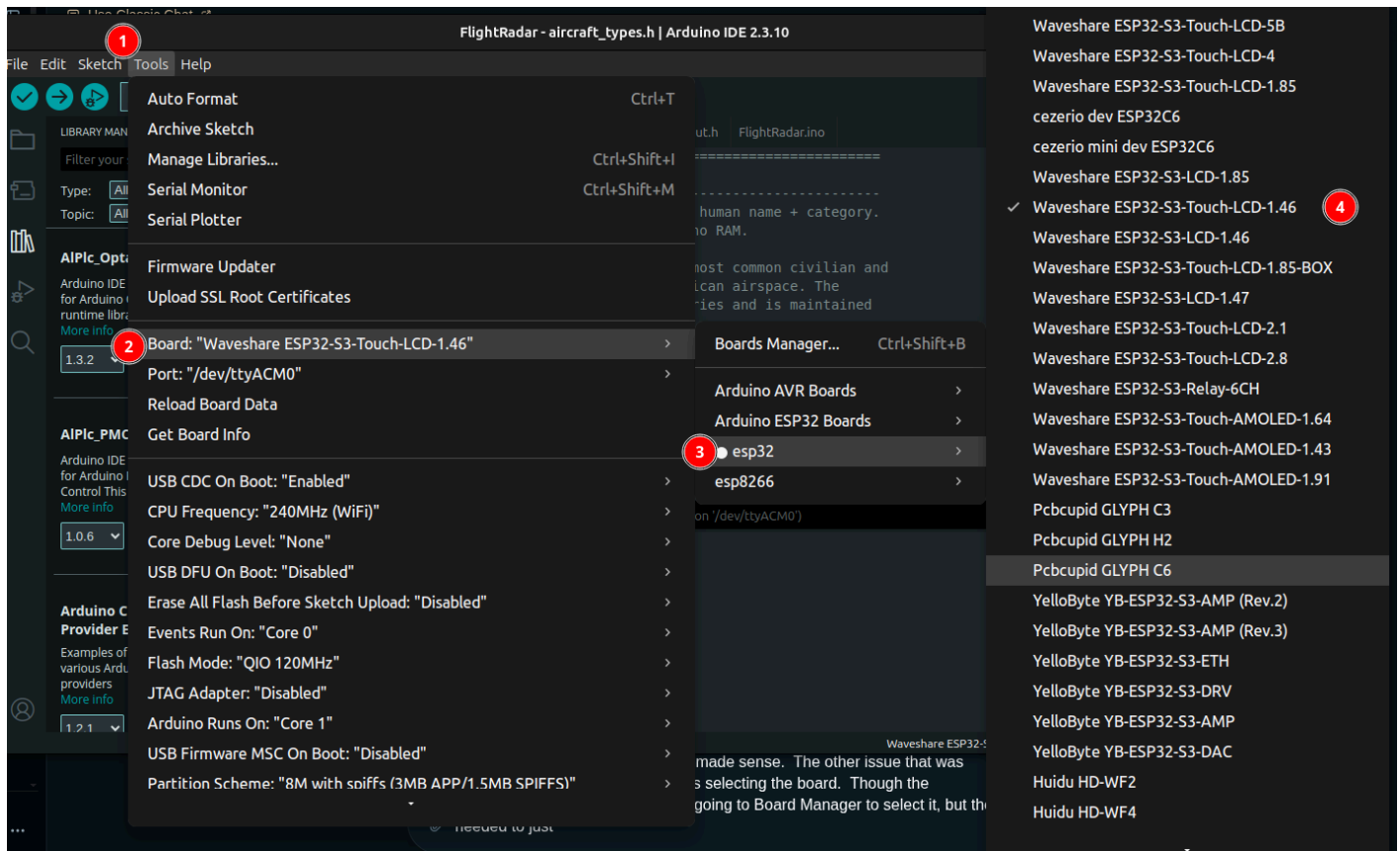
Part 5: Configure Board Settings

Connect your board to the computer with a USB-C **data cable** (not a charge-only cable).

Select the Board

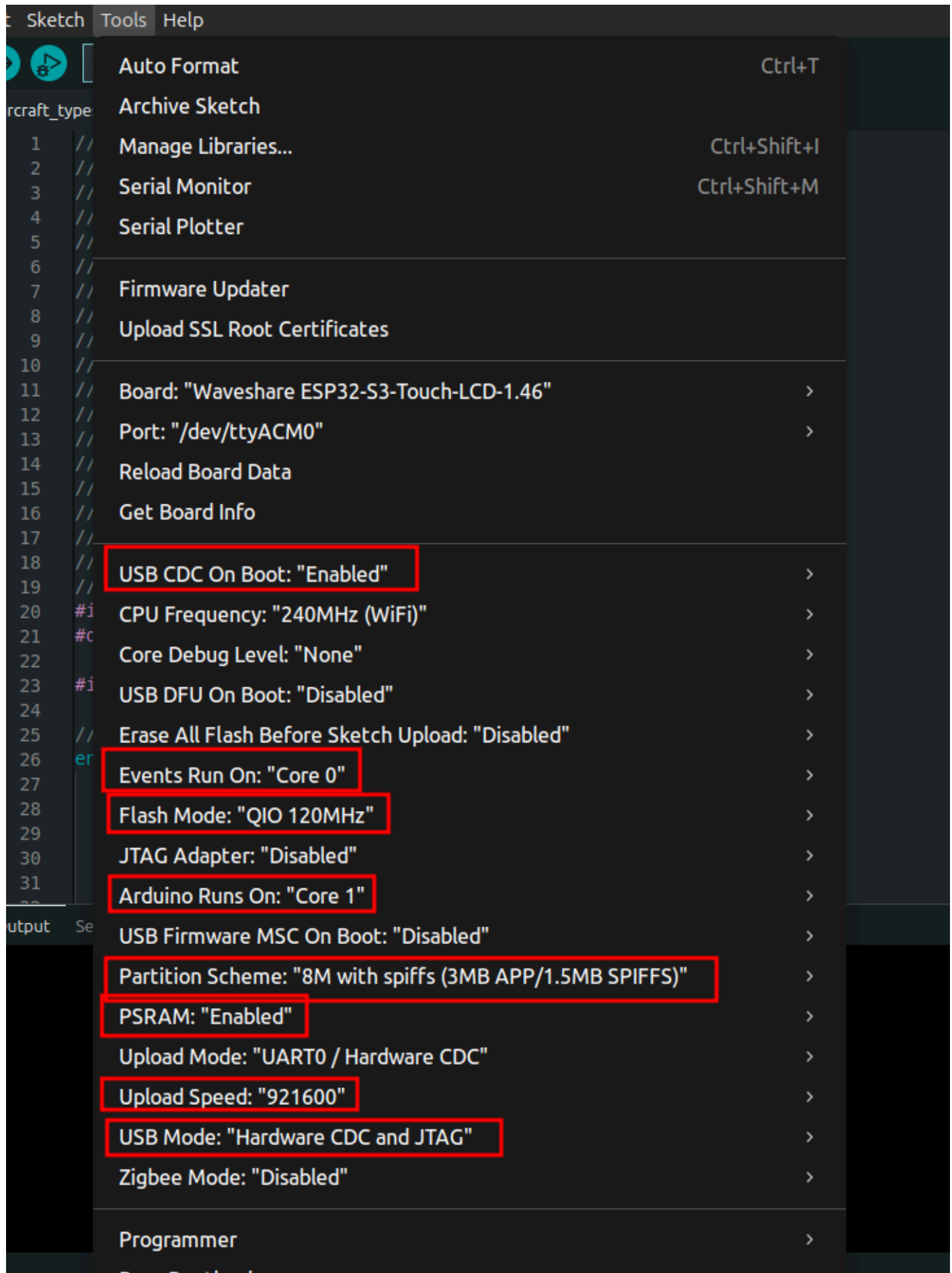
Go to **Tools** menu and set **exactly** these options:

Board:	Wavesheare ESP32-S3-Touch-LCD-1.46
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```

USB CDC On Boot:      Enabled
Events Run On:        Core 0
Flash Mode:           QIO 120MHz
Arduino Code Runs On: Core 1
Partition Scheme:     8M with spiffs (3MB APP/1.5MB SPIFFS)
PSRAM:                Enabled      ← CRITICAL!
Upload Speed:         921600
USB Mode:             Hardware CDC and JTAG
  
```



⚠ **Critical:** PSRAM must be set to **"Enabled"** (not "Disabled"). The display framebuffer requires PSRAM. If PSRAM is disabled, the screen will stay black.

⚠ **Critical:** Partition Scheme **MUST** have **"SPIFFS"**

Select the Port

Go to **Tools** → **Port** and select:

- **Windows:** COM3 or similar (may show "USB-SERIAL")
- **Mac:** /dev/cu.usbserial-XXXX or /dev/cu.wchusbserial-XXXX
- **Linux:** /dev/ttyUSB0 or similar

If no port appears:

- Try a different USB cable (must be data cable, not charge-only)
- Try holding the **BOOT** button and press the **PWR/RST** button (release both buttons after pressing the PWR/RST button)
- Install CP210x USB driver (Windows): <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>
- On Mac, wait 30 seconds after plugging in—ports take time to appear

Part 6: Upload the Code

Click the → **(Upload)** button or select **Sketch** → **Upload**

If Upload Fails

Error: "Failed to connect to ESP32"

Use the BOOT button method:

1. Hold down the **BOOT** button on the board (small button labeled "BOOT")
2. While holding, click **Upload** in Arduino IDE
3. Keep holding until you see "Connecting..." in the output window
4. Release the BOOT button
5. If it still fails, try again

Error: "A fatal error occurred: Failed to write to target RAM"

- Press the **RST** button on the board
 - Might be labeled **PWR**
- Try upload again with BOOT button held

Success: You should see:

```
Leaving...  
Hard resetting via RTS pin...
```

Part 7: Verify It Works

Check the Serial Monitor

1. Open **Tools** → **Serial Monitor** (or press Ctrl+Shift+M / Cmd+Shift+M)
2. Set baud rate to **115200** (bottom right dropdown)
3. Press the **RST** (or **PWR**) button on your board

You should see:

```
[Boot] FlightRadar starting...  
[Boot] Waveshare ESP32-S3-Touch-LCD-1.46C  
[Touch] SPD2010 reset complete  
[IMU] QMI8658 ready  
[FS] LittleFS ready – XXXX KB free  
[LCD] panel + canvas ready  
[WiFi] connected – IP=192.168.X.XXX  
[mDNS] flightradar.local  
[OTA] ready  
[HTTP] web server up on port 80  
[Boot] ready
```

Check the Display

The round LCD should show:

"SETUP" in the center with a WiFi name like:

```
FlightRadar-A1B2
```

If you see this, **success!** Proceed to WiFi configuration in the [Student Guide](#).

Troubleshooting

Problem	Solution
No port appears	Try different USB cable; install CP210x driver (Windows); wait 30s (Mac)

Problem	Solution
No port appears	With BOOT button held, press RST/PWR
"Failed to connect"	Hold BOOT button during upload
"Canvas->begin() FAILED"	Check PSRAM is set to Enabled in Tools menu
Upload hangs at "Connecting..."	With BOOT button held, press RST/PWR
Display stays black	Press RST/PWR button; verify PSRAM setting
"Library not found"	Check spelling; install via Library Manager
Serial Monitor shows garbage	Set baud rate to 115200, not 9600
Touch doesn't work	Restart board; SPD2010 needs a few seconds after boot

Next Steps

- **Students:** Return to [Student Guide](#) for WiFi configuration
- **Instructors:** See [Leader Guide](#) for teaching notes and troubleshooting tips

Advanced: Flashing Without USB (OTA)

Once the board is running and connected to WiFi, you can flash updates wirelessly:

1. In Arduino IDE: **Tools** → **Port**
2. Look for **flightradar at 192.168.X.XXX** (network port)
3. Select it
4. Click Upload as normal

No USB cable needed! Requires Arduino IDE and board on same WiFi network.